

Cesai Li

Curriculum Vitae

Contact Information

Office Address: Department of Mathematics and Statistics
Boston University
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Education

Ph.D. in Mathematics, Boston University 2024–present

Bachelor of Science in Mathematics, University of British Columbia 2020–2024
Graduated with Distinction

Research Interests

Mathematical physics, quantum field theory, factorization algebra, algebraic structures of quantum observables, applications of category theory and homological algebra in mathematical physics

Research Experience and Projects

- **Research Assistant**, University of Saskatchewan 2023 Summer
Advisor: Alex Weekes
– Topic: *On the monopole formula and its generalizations*

Teaching Experience

- **Teaching Fellow** for CAS MA 115C Statistics I, BU 2026 Spring
- **Teaching Fellow** for CAS MA 242C Linear Algebra, BU 2025 Fall
- **Instructor of Record** for CAS MA 116 Statistics II, BU 2025 Summer 2
- **Instructor of Record** for CAS MA 116 Statistics II, BU 2025 Summer 1
- **Teaching Assistant** for MATH 100 Differential Calculus, UBC 2022 Fall

Honours and Awards

- Dean's Fellowship, Boston University 2024 Fall–2025 Spring
- The Canadian Federation of University Women Gaming Education Award 2023
- Dean's Honour List, University of British Columbia 2021

Community Involvement

- **Directed Reading Program Mentor**, Boston University 2025 Spring
Mentee: Zhengkai Li
– Topic: *Complete classification of finitely generated abelian groups*

Talks and Presentations

- Research group seminar Nov 12, 2025
Topic: *Infinite category constructions: simplicial sets, quasicategories, and simplicial categories*
- Research group seminar Sep 17, 2025
Topic: *Computations with the Jones polynomial*

Conferences and Workshops Attended

- New England Algebraic Topology and Mathematical Physics Seminar NOV 15, 2025
– A miniconference in topology and quantum field theory at Boston University, Boston, MA
- Homotopical algebra in geometry, topology, and physics JUN 20-24, 2025
at Northwestern University, Evanston, IL
- Physical Mathematics of QFT Summer School 2025 JUN 09-13, 2025
at University of Massachusetts Amherst, Amherst, MA
- GTA: Philadelphia 2025 10th Annual Graduate Student Conference MAY 30-JUN 01, 2025
in Algebra, Geometry, and Topology at Temple University, Philadelphia, PA
- New England Algebraic Topology and Mathematical Physics Seminar NOV 2–3, 2024
– A miniconference in topology and quantum field theory at Providence College, Providence, RI